Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-4 | Summer-2023

Course Code	:	2101CS403	**Date**	:	05-04-2023
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Discuss role of OS as a resource manager.	**4**
	(B)	Explain the features of Time-sharing system.	**3**
		OR	
		Explain monolithic system.	
	(C)	Discuss different services provided by operating system.	**7**
		OR	
		Discuss different types of operating system.	
Q.2	**(A)**	Define: process, thread and context switching.	**4**
	(B)	Describe process control block (PCB) with diagram.	**3**
		OR	
		Describe Process Life Cycle with diagram.	
	(C)	Calculate the average waiting time using the FCFS and SJF scheduling algorithm for the processes P1, P2, P3, P4 with burst time is 21, 3, 6 and 2 respectively, arrives for execution in the same order, with arrival time 0 also draw GANTT chart.	**7**
		OR	
		Calculate the average turnaround time and average waiting time for Round Robin (quantum time = 3) and Priority Scheduling algorithm. Consider Five Processes P1 to P5 arrived at same time. They have estimated running time 10, 2, 6, 8 and 4 seconds, respectively. Their Priorities are 3, 2, 5, 4 and 1, respectively with 5 being highest Priority.	
Q.3	**(A)**	Define: Mutual Exclusion, Critical Section, Race Condition	**4**
	(B)	Describe is Deadlock? List the conditions that lead to deadlock.	**3**
		OR	
		Discuss Banker's algorithm for deadlock avoidance.	
	(C)	Define Semaphore. Discuss Readers-Writers Problem using Semaphore.	**7**

OR

Define Semaphore. Discuss Bounded Buffer Producer Consumer Problem using Semaphore.

Q.4	**(A)**	Explain how free space can be managed by OS.	**4**
	(B)	Explain paging with the help of diagram.	**3**

OR

Explain page table with the help of diagram.

	(C)	Calculate the page fault rates for below reference string in case of FIFO and Optimal page replacement algorithm. Assume the memory size is 4-page frames and all frames are initially empty. 0,2,1,6,4,0,1,0,3,1,2,1.	**7**

OR

Demonstrate how would the first-fit, best-fit and worst-fit algorithms places processes of size 115 KB, 500KB, 358KB, 200KB and 375KB(in order) in to the six Partition of 300KB, 600KB, 350KB, 200KB, 750KB and 125KB(in order). Which algorithm is efficient for the use of memory?

Q.5	**(A)**	Explain different types of operations performed on the file.	**4**
	(B)	Discuss DMA.	**3**

OR

Discuss RAID.

	(C)	Calculate the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for FCFS and SSTF disk scheduling algorithm. Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order.	**7**

OR

Calculate the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for LOOK and SCAN disk scheduling algorithm. Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order.

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